A Note on the $k = 5$ Case of the Min-Sum Boundary Objective for Hypergraph Partitions

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Abstract

Let $G = (V, E)$ be a hypergraph, and let

$$d_G(U) = |\delta_G(U)| = \big|\{e \in E : e \cap U \neq \emptyset,\ e \cap (V \setminus U) \neq \emptyset\}\big|.$$

This note considers the fixed-k problem of minimizing

$$\sum_{i=1}^{k} d_G(V_i)$$

over all k-partitions $(V_1, \ldots, V_k)$ of V into nonempty parts. The main mathematical observation is that d_G is a normalized symmetric submodular function. Therefore the case $k = 5$ is exactly a symmetric-submodular 5-partition instance, and is polynomial-time solvable by the known reduction from symmetric-submodular 5-partition to general submodular 4-partition together with the polynomial-time algorithm of Hirayama, Liu, Makino, Shi, and Xu.

The status is as follows. This gives a complete proof for the isolated case $k = 5$, but it is not a solution of the currently stated open problem in the published paper of Beideman, Chandrasekaran, and Wang: that paper already records polynomial-time solvability for $k \leq 5$ and leaves $k \geq 6$ open. Thus the note is best viewed as a rigorous explanation of the known $k = 5$ progress, not as new progress on the remaining $k \geq 6$ problem.

1 Status relative to the open problem

Beideman, Chandrasekaran, and Wang study hypergraph k-partitioning problems. In addition to the main objectives in their paper, they discuss the related min-sum boundary variant

$$\min\left\{\sum_{i=1}^{k} |\delta_G(V_i)| \ :\ (V_1, \ldots, V_k) \text{ is a } k\text{-partition of } V\right\}. \tag{1}$$

The usual hypergraph k-cut objective counts each hyperedge crossing a k-partition once. In contrast, (1) counts a crossing hyperedge once for every part that it meets.

An earlier/preliminary formulation asked for the complexity of this min-sum boundary variant for fixed $k \geq 5$ [1]. The published version records that the variant is polynomial-time solvable for $k \leq 5$ and says that the remaining unresolved case is $k \geq 6$ [2, Section 7]. The purpose of this note is to give the short proof of the $k = 5$ statement in a self-contained way, using the known symmetric-submodular 5-partition theorem.

Consequently, the contribution here is precisely this:

$$\boxed{\text{a complete proof for } k = 5, \text{ but not a solution of the remaining } k \geq 6 \text{ open problem.}}$$

2 Definitions and relation to nearby objectives

A *k-partition* of V is an ordered tuple $(V_1, \dots, V_k)$ of pairwise disjoint nonempty subsets of V whose union is V. The ordering is immaterial for the objective, but using ordered tuples avoids quotient notation.

For $U \subseteq V$, define

$$\delta_G(U) = \{e \in E : e \cap U \neq \emptyset \text{ and } e \cap (V \setminus U) \neq \emptyset\}, \qquad d_G(U) = |\delta_G(U)|.$$

For a weighted hypergraph with nonnegative weights $c : E \to \mathbb{R}_{\geq 0}$, the corresponding weighted boundary function is

$$d_{G,c}(U) = \sum_{e \in \delta_G(U)} c(e).$$

The unweighted case is obtained by setting $c(e) = 1$ for every $e \in E$.

For a partition $\mathcal{P} = (V_1, \dots, V_k)$ and hyperedge e, write

$$r_e(\mathcal{P}) = \bigl|\{i \in \{1, \dots, k\} : e \cap V_i \neq \emptyset\}\bigr|.$$

Then

$$\sum_{i=1}^{k} d_G(V_i) = \sum_{e \in E} \psi\bigl(r_e(\mathcal{P})\bigr), \qquad \psi(1) = 0, \quad \psi(r) = r \text{ for } r \geq 2. \tag{2}$$

Thus the min-sum boundary objective is not the usual hypergraph k-cut objective, which would use the contribution 0 for $r = 1$ and 1 for every $r \geq 2$. It is also distinct from the normalized-coverage objective, which charges $r - 1$ for a hyperedge meeting r parts. For graphs, every crossing edge meets exactly two parts, so (1) is just twice the usual graph k-cut objective. For hypergraphs, the objectives differ.

A set function $f : 2^V \to \mathbb{R}$ is *submodular* if

$$f(A) + f(B) \geq f(A \cap B) + f(A \cup B) \qquad \text{for all } A, B \subseteq V.$$

It is *symmetric* if $f(A) = f(V \setminus A)$ for all $A \subseteq V$, and *normalized* if $f(\emptyset) = f(V) = 0$.

3 The boundary function is symmetric submodular

We assume throughout that hyperedges are nonempty. Empty hyperedges never cross a cut and may simply be deleted; this convention is immaterial for the optimization problem.

Lemma 3.1. *For every hypergraph $G = (V, E)$ with nonempty hyperedges, the function $d_G : 2^V \to \mathbb{Z}_{\geq 0}$ is normalized, symmetric, and submodular. More generally, if $c : E \to \mathbb{R}_{\geq 0}$, then $d_{G,c}$ is normalized, symmetric, and submodular.*

Proof. Normalization and symmetry are immediate from the definition:

$$d_G(\emptyset) = d_G(V) = 0, \qquad d_G(U) = d_G(V \setminus U).$$

It remains to prove submodularity. For a hyperedge $e \in E$, define

$$\chi_e(U) = \begin{cases} 1, & e \cap U \neq \emptyset \text{ and } e \cap (V \setminus U) \neq \emptyset, \\ 0, & \text{otherwise.} \end{cases}$$

Then $d_G(U) = \sum_{e \in E} \chi_e(U)$, so it is enough to show that each χ_e is submodular.

Let

$$a_e(U) = \mathbf{1}[e \cap U \neq \emptyset].$$

Since e is nonempty,

$$\chi_e(U) = a_e(U) + a_e(V \setminus U) - 1. \tag{3}$$

The function a_e is submodular. Indeed, for $X, Y \subseteq V$, if e meets both X and Y, then the left-hand side $a_e(X) + a_e(Y)$ is 2 and the desired inequality is clear; if e fails to meet one of X, Y, then e fails to meet $X \cap Y$, and the inequality is again immediate. Equivalently, a_e is a rank-one coverage function.

The complement transform $U \mapsto a_e(V \setminus U)$ is also submodular: applying submodularity of a_e to $V \setminus X$ and $V \setminus Y$ gives

$$a_e(V \setminus X) + a_e(V \setminus Y) \geq a_e(V \setminus (X \cap Y)) + a_e(V \setminus (X \cup Y)).$$

By (3), χ_e is therefore submodular. A nonnegative linear combination of submodular functions is submodular, so both d_G and $d_{G,c}$ are submodular. □

4 The external submodular-partition theorem

We use the following known result as a black box.

Theorem 4.1 (Hirayama–Liu–Makino–Shi–Xu; symmetric-submodular corollary)**.** *Let $f : 2^V \to \mathbb{R}$ be a symmetric submodular function on an n-element ground set, given by an evaluation oracle. Then a 5-partition $(V_1, \dots, V_5)$ minimizing*

$$\sum_{i=1}^{5} f(V_i)$$

can be found in polynomial time. More quantitatively, using a minimum 4-partition algorithm for general submodular functions, one obtains an $O(n^{14}\tau(n))$ bound, where $\tau(n)$ denotes the time for the submodular-function-minimization oracle used in that framework.

Citation and explanation. Hirayama, Liu, Makino, Shi, and Xu prove a polynomial-time exact algorithm for minimum 4-partition of a general submodular function [4]. They also state the consequence that a minimum 5-partition of a symmetric submodular function can be found in $O(n^{14}\tau(n))$ time. The consequence follows from the standard reduction, cited there, that minimum k-partition for symmetric submodular functions can be solved using n^{2k-2} calls to minimum $(k-1)$-partition for general submodular functions. Setting $k = 5$ invokes the general submodular 4-partition algorithm. □

5 The $k = 5$ theorem

Theorem 5.1. *For explicitly represented unweighted hypergraphs, the problem*

$$\min \left\{ \sum_{i=1}^{5} |\delta_G(V_i)| \, : \, (V_1, \dots, V_5) \text{ is a 5-partition of } V \right\}$$

is solvable in polynomial time. The same is true for explicitly represented hypergraphs with nonnegative rational weights.

Proof. If $|V| < 5$, no 5-partition exists. Assume $|V| \geq 5$.

By Lemma 3.1, the function d_G is symmetric submodular. The displayed optimization problem is exactly the minimum 5-partition problem for the symmetric submodular function $f = d_G$. Therefore Theorem 4.1 gives a polynomial-time algorithm in the value-oracle model.

It remains only to note that, for an explicit hypergraph, the value oracle for d_G is polynomial-time. Given $U \subseteq V$, one scans the incidence list of each hyperedge and tests whether the hyperedge has at least one vertex in U and at least one vertex in $V \setminus U$. Thus each evaluation of $d_G(U)$ takes time polynomial in the input representation size. Combining this value oracle with the oracle algorithm from Theorem 4.1 gives a polynomial-time algorithm in the explicit hypergraph input model.

For nonnegative rational weights, replace d_G by $d_{G,c}$. Lemma 3.1 still applies, and exact rational arithmetic gives polynomial bit complexity under the standard encoding of rational weights. $\square$

Corollary 5.2. *For $k = 5$, the edge-contribution objective*

$$\sum_{e \in E} \psi\big(r_e(\mathcal{P})\big), \qquad \psi(1) = 0, \quad \psi(r) = r \text{ for } r \geq 2,$$

can be minimized exactly in polynomial time over all 5-partitions $\mathcal{P}$ of V.

Proof. This is just Theorem 5.1 rewritten using identity (2). $\square$

6 Why this does not solve the remaining $k \geq 6$ problem

The proof above is sharp with respect to the currently available black box. It uses the fact that symmetric-submodular 5-partition reduces to general submodular 4-partition, and general submodular 4-partition is known to be polynomial-time solvable. For $k = 6$, the same route would require a polynomial-time algorithm for general submodular 5-partition, or a different argument exploiting special structure of hypergraph boundary functions.

A natural attempt is to fix one part A of the desired partition and then solve a residual problem on $R = V \setminus A$. This does not preserve the symmetry needed by Theorem 4.1. Indeed, for a partition $(R_1, \ldots, R_{k-1})$ of R, the residual terms have the form

$$\sum_{j=1}^{k-1} d_G(R_j),$$

where d_G is still evaluated as a cut function in the original ground set V, not in R. The restriction

$$g_A(S) = d_G(S), \qquad S \subseteq R,$$

is submodular, but it need not be symmetric with respect to the ground set R.

For example, let $V = \{a, s, t\}$, let $A = \{a\}$, let $R = \{s, t\}$, and let $E = \{\{a, s\}\}$. Then

$$g_A(\{s\}) = d_G(\{s\}) = 1, \qquad g_A(R \setminus \{s\}) = g_A(\{t\}) = d_G(\{t\}) = 0.$$

Thus the residual function is not symmetric on R. This illustrates why the $k = 5$ argument cannot simply be iterated to obtain $k = 6$.

Equivalently, once a part A is fixed, hyperedges touching A induce coverage-type residual terms, while hyperedges disjoint from A continue to induce boundary-type terms. The mixture remains submodular but is no longer symmetric in general. The known exact polynomial-time theorem for arbitrary submodular partitioning currently reaches four parts in the black-box route used here, which is exactly enough for symmetric five-partition and not enough for symmetric six-partition.

7 Conclusion

The min-sum boundary objective for a hypergraph is the submodular partition objective for the hypergraph boundary function d_G. Since d_G is symmetric submodular, the known polynomial-time algorithm for symmetric-submodular 5-partition gives an immediate polynomial-time algorithm for the hypergraph objective when $k = 5$.

This is a complete solution for the isolated $k = 5$ case. It is not a solution to the currently remaining problem in the published Beideman–Chandrasekaran–Wang paper, because the published paper already incorporates the $k \le 5$ solvability and leaves $k \ge 6$ open. Any full fixed-k resolution beyond this point would need a new idea, such as a polynomial-time algorithm for general submodular 5-partition or a proof that the additional structure of hypergraph boundary functions can be exploited for $k \ge 6$.

References

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